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CSA0609- Design and Analysis of Algorithms

2 Common Elements Between Two Arrays and find its completity

AIM

To write a Python program to find the common elements between two arrays**.**

ALGORITHM

1. Start.
2. Read two arrays.
3. Compare the elements of both arrays.
4. If an element is present in both, store it.
5. Display the common elements.
6. Stop.

PSEUDOCODE

```
Begin
  common ← empty list
  for each element in A do
    if element is in B then
      add element to common
    end if
  end for
  print common
End
```

PYTHON CODE

main.py Visualize Run

```
A = list(map(int, input("Enter first array: ").split()))
B = list(map(int, input("Enter second array: ").split()))

common = []

for x in A:
    if x in B and x not in common:
        common.append(x)

print("Common Elements:", common)
```

OUTPUT

Output

```
Enter first array: 10 20 30 40 50
Enter second array: 30 40 60 70 80
Common Elements: [30, 40]
```

COMPLEXITY ANALYSIS

Let:

- n = number of elements in first array
- m = number of elements in second array

Time Complexity = O(n × m)

Space Complexity = O(min(n, m))

RESULT

Thus, the program successfully finds the common elements between two arrays with O(n × m) time complexity and O(min(n, m)) space complexity.